

Project Dispatch

The Proactive Transit & Logistics Orchestrator

Executive Summary

Most digital assistants are fundamentally reactive—they wait for a wake word, search an API, and recite a result. Furthermore, most travel and logistics applications only alert users to disruptions *after* a delay has cascaded through the official channels. Project Dispatch reimagines the "Jarvis" paradigm for the physical world.

Dispatch is an autonomous, localized orchestration engine that monitors raw transit telemetry in the background. By ingesting unstructured data streams—such as live train GPS coordinates and aircraft state vectors—the system anticipates scheduling failures before they happen. When a disruption is detected, it does not wait to be asked; it proactively synthesizes an auditory briefing and executes automated mitigations (such as drafting cancellation emails or rescheduling calendar events).

Hackathon Track Alignment

NVIDIA Nemotron

Beyond the Chatbot:

Nemotron operates strictly as a silent, predictive logic engine. It takes in raw, noisy telemetry logs, analyzes anomalous behaviors (e.g., a train sitting stationary for too long), and outputs structured JSON action payloads.

ElevenLabs

Out Loud: Brings the "Jarvis" aesthetic to life. Instead of relying on push notifications, Dispatch proactively speaks to the room. It generates spatial audio briefings only when actionable intervention is required.

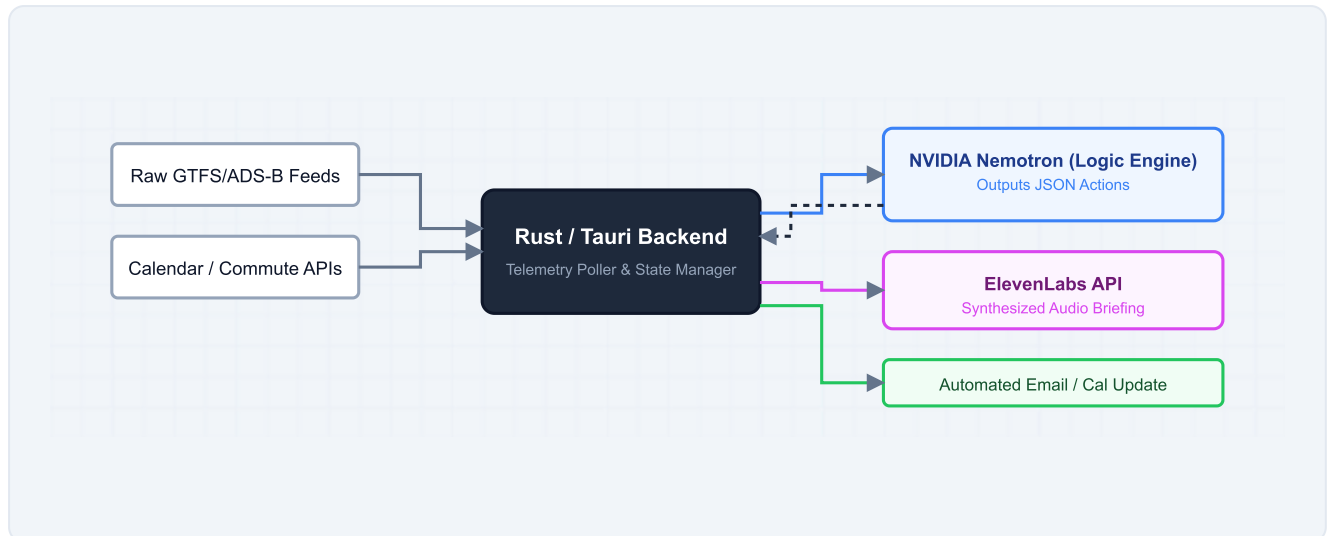
Afore Capital

Seed Round: A proactive assistant that defends a user's schedule has massive commercial appeal for enterprise travelers and logistics managers, representing a highly fundable B2B or B2C SaaS product.

Core Concept: Move away from LLMs as conversational conversationalists. Use them as unstructured-to-structured data compilers that drive automated, real-world physical logic.

System Architecture

The system is designed to be lightweight, running locally as a desktop application to maintain the illusion of an integrated personal assistant, rather than a web dashboard.



Technical Stack Details

- **Frontend App Wrapper:** Tauri (Rust + React/Vite). Provides a lightweight system tray application that minimizes to the background. This ensures it feels like a native assistant, not a browser tab.
- **State Engine (Backend):** A Python or Rust polling daemon that fetches data from the configured transit APIs on a 30-to-60-second cron cycle.
- **The Intelligence Layer:** NVIDIA Nemotron. The backend feeds Nemotron a system prompt containing the user's itinerary alongside raw transit logs. Nemotron is prompted strictly to return a JSON object (e.g., `{"status": "disrupted", "action": "synthesize_warning", "message": "..."}`).
- **Voice Synthesis:** ElevenLabs API. When Nemotron flags a disruption, the backend sends the generated warning text to ElevenLabs, streaming the audio directly through the user's desktop speakers.

Telemetry & Data Sources (Pittsburgh Focus)

To execute the proactive forecasting element of this project, Dispatch requires access to live, raw telemetry. The following data sources provide localized coverage for the Greater Pittsburgh region.

Domain	Data Source & API	Implementation Strategy
Local Transit (Buses & Light Rail)	PRT GTFS-RT via WPRDC	The Western Pennsylvania Regional Data Center (WPRDC) hosts the GTFS Realtime feeds for Pittsburgh Regional Transit. You will poll the <code>VehiclePositions</code> protobuf feed. By tracking the latitude, longitude, and timestamp of specific bus routes, Nemotron can detect unannounced traffic stalls before PRT issues a service alert.
Regional Rail (Intercity Commuting)	Amtrak Live Telemetry (Track Your Train)	Amtrak does not have a public API. You will need to scrape the live JSON feed powering their "Track Your Train" web map (or use a wrapper like Apify). This tracks the precise speed and location of the <i>Pennsylvanian</i> (Train 42/43). If the speed stays at 0 mph for 15+ minutes between stations, Dispatch triggers a rescheduling event.
Air Logistics (Pittsburgh Int'l - PIT)	OpenSky Network (ADS-B Data)	The OpenSky Network provides free access to live airspace information for non-commercial use. Instead of relying on airline departure boards, poll the raw state vectors (position, velocity, altitude) of aircraft bounding for PIT. Detecting holding patterns allows Dispatch to proactively alert the user to delay their drive to the airport.

Development Milestones for Hackathon

- **Hour 1-4:** Establish basic API polling. Get PRT GTFS-RT vehicle positions parsing in the terminal.
- **Hour 5-10:** Connect Nemotron. Feed it the parsed transit data and prompt engineer it to reliably output the JSON action schema.
- **Hour 11-16:** Integrate ElevenLabs. Wire the JSON output to trigger spatial audio generation on the desktop.
- **Hour 17-24:** Refine the Tauri UI/System Tray and record the demo. Focus the pitch on the shift from "reactive chatbot" to "proactive logistics engine."